Cagrilintide–semaglutide (CagriSema) as an add-on to basal insulin in adults with type 2 diabetes (REIMAGINE 3): a randomised, double-blind, placebo-controlled, multicentre, phase 3 study

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Summary

Background Basal insulin treatment for type 2 diabetes often results in inadequate glycaemic control and is associated with weight gain and increased risk of hypoglycaemia. We aimed to compare the efficacy and safety of a once per week combination of cagrilintide with semaglutide (CagriSema) versus placebo as an add-on to basal insulin in individuals with type 2 diabetes.

Methods This double-blind, parallel-group, randomised, controlled, phase 3a study (REIMAGINE 3) was done at 46 centres (university hospitals, health-care centres, research centres, and other clinical trial sites) in six countries (the USA, China, Japan, Serbia, Slovakia, and South Africa). Adults with type 2 diabetes (glycated haemoglobin [HbA_{1c}] 7·0–10·5%) receiving stable once per day basal insulin with or without metformin were randomly assigned (2:2:1:1) to once per week subcutaneous cagrilintide 2·4 mg plus semaglutide 2·4 mg (hereafter cagrilintide–semaglutide [2·4 mg each]) or cagrilintide 1·0 mg plus semaglutide 1·0 mg (hereafter cagrilintide–semaglutide [1·0 mg each]) or dose-matched placebo (2·4 mg plus 2·4 mg or 1·0 mg plus 1·0 mg) for 40 weeks. Randomisation was stratified by HbA_{1c} of less than 8·5% at screening (yes or no) and country (Japan; yes or no). Participants, care providers, investigators, and outcome assessors were masked within each dose level to treatment allocation, with active treatments visually identical to placebo. The primary endpoint was mean HbA_{1c} change (percentage points) from baseline to week 40 in all participants randomly assigned to treatment; secondary endpoints included change in bodyweight and safety, including hypoglycaemia. This trial was registered with ClinicalTrials.gov (NCT06323161) and is complete.

Findings Between March 26 and Nov 29, 2024, we screened 340 individuals and 274 were included and randomly assigned to cagrilintide–semaglutide (2·4 mg each; n=90), cagrilintide–semaglutide (1·0 mg each; n=93), or placebo (pooled; n=91). Mean baseline HbA_{1c} was 8·8% (SD 1·0), 159 (58%) participants were male, and 115 (42%) were female. Mean HbA_{1c} reductions were significantly greater with cagrilintide–semaglutide (2·4 mg each –2·33% [SE 0·08] and 1·0 mg each –2·10% [0·08]) versus placebo (–0·66% [0·11]) at week 40, using the efficacy estimand (estimated treatment difference for cagrilintide–semaglutide [2·4 mg each] *vs* placebo –1·68 percentage points [95% CI –1·95 to –1·41], p<0·0001; for cagrilintide–semaglutide [1·0 mg each] *vs* placebo –1·44 percentage points [95% CI –1·71 to –1·17], p<0·0001). Cagrilintide–semaglutide provided bodyweight reductions of 10–12%. Adverse events were reported by 72 (80%) of 90 participants receiving cagrilintide–semaglutide (2·4 mg each), 66 (71%) of 93 receiving cagrilintide–semaglutide (1·0 mg each), and 65 (71%) of 91 receiving placebo, and were mostly mild or moderate gastrointestinal disorders. No severe hypoglycaemia was reported. There was one death in the cagrilintide–semaglutide (1·0 mg each) group not related to treatment (due to malignancy).

Interpretation Cagrilintide–semaglutide at doses of 2·4 mg each and 1·0 mg each met the primary endpoint, with statistically significant and clinically relevant HbA_{1c} reductions versus placebo when added to basal insulin-treated type 2 diabetes. These reductions were accompanied by robust bodyweight reduction and no additional risk of hypoglycaemia. The safety profile was consistent with that of the GLP-1 receptor agonist class and previous safety data for cagrilintide. Findings support the use of cagrilintide–semaglutide as an add-on to once per day basal insulin to significantly improve glycaemic control.

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Research in context

Evidence before this study

We searched PubMed on Feb 17, 2026, for studies in any language with the terms "type 2 diabetes" AND ("semaglutide" OR "glucagon-like peptide-1 receptor agonist" OR "GLP-1") AND ("cagrilintide" OR "amylin analog*") AND ("basal insulin" OR "insulin glargine" OR "insulin detemir" OR "insulin degludec"), and limited our selection to randomised controlled clinical trials. This search yielded no results; so far, no phase 3 studies have investigated cagrilintide–semaglutide as an add-on therapy to basal insulin in people with type 2 diabetes. We repeated the search, omitting ("basal insulin" OR "insulin glargine" OR "insulin detemir" OR "insulin degludec") from the search terms. Cagrilintide–semaglutide (2·4 mg each) was investigated in a phase 2 study in people with type 2 diabetes and overweight or obesity on metformin with or without an SGLT2 inhibitor. Cagrilintide–semaglutide resulted in clinically relevant improvements in glycated haemoglobin and significantly greater bodyweight reduction than either cagrilintide or semaglutide alone. In the phase 3a REDEFINE 2 study for weight management in people with type 2 diabetes and overweight or obesity receiving lifestyle intervention or glucose-lowering agents other than GLP-1 receptor agonists or insulin, cagrilintide–semaglutide (2·4 mg each) was associated with significantly greater reductions in bodyweight than placebo, and significantly greater proportions of participants met bodyweight reduction targets of 5–20%. This was accompanied by improvements in a range of glycaemic and cardiometabolic endpoints. Cagrilintide–semaglutide was well tolerated in both studies, with a safety profile that was generally consistent with previous studies of cagrilintide and the GLP-1 receptor agonist class.

Added value of this study

REIMAGINE 3 extends the evaluation of cagrilintide–semaglutide to individuals receiving basal insulin, a clinically complex population in whom late-stage disease is common, thereby addressing a key evidence gap regarding the role of dual amylin and GLP-1 receptor agonism in this treatment setting. Whereas REIMAGINE 1 investigated efficacy across less intensive treatment settings and REIMAGINE 2 assessed efficacy versus active comparators, REIMAGINE 3 provides evidence in individuals for whom treatment intensification with insulin is often associated with weight gain, hypoglycaemia risk, and therapeutic inertia. Collectively, the REIMAGINE trials provide a comprehensive assessment of cagrilintide–semaglutide across the type 2 diabetes treatment continuum, supporting its potential to address unmet needs related to glycaemic regulation, weight management, metabolic control, and treatment burden across diverse patient populations and lines of therapy.

Implications of all the available evidence

Treatment intensification in type 2 diabetes in those receiving basal insulin therapy is often accompanied by risk of weight gain and hypoglycaemia, and glycaemic targets are often not achieved. The results of REIMAGINE 3 support cagrilintide–semaglutide as an effective option for advancing therapy in inadequately controlled, basal insulin-treated type 2 diabetes to achieve meaningful near-normoglycaemia. In addition, cagrilintide–semaglutide was associated with robust bodyweight reduction, with no increase in the risk of hypoglycaemia observed in this study.

Introduction

A substantial proportion of people with basal insulin-treated type 2 diabetes exhibit suboptimal glycaemic control, which could contribute to an increased risk of complications, including chronic kidney disease and cardiovascular disease.[1–3] Insulin therapy is associated with an increased risk of hypoglycaemia,[4] and insulin-associated weight gain might further hinder attainment of glycaemic treatment targets.[5,6] As such, this population might benefit from the introduction of novel therapies that optimise glycaemic control and weight management and reduce cardiorenal risk in people with type 2 diabetes, as recommended in current clinical guidelines.[7–9]

GLP-1-based therapies have been associated with improvements in glycaemic control, weight management, and cardioprotective effects in people with type 2 diabetes.[10–12] In randomised studies of basal insulin-treated participants with type 2 diabetes who do not meet treatment targets, GLP-1-based therapies have also been shown to improve glycaemic control and induce greater bodyweight reductions than prandial insulin and intensified basal insulin.[13,14] Use of prandial insulin is frequently accompanied by adverse events including further weight gain and an increased risk of hypoglycaemia.[4]

CagriSema (Novo Nordisk, Bagsværd, Denmark) is a fixed-dose combination of cagrilintide, a long-acting amylin receptor agonist, and the GLP-1 receptor agonist semaglutide. Cagrilintide has been shown to reduce food intake and induce bodyweight reduction in phase 2 and phase 3 clinical studies, and has shown glucose-lowering properties.[15,16] Semaglutide is indicated for glycaemic control in people with type 2 diabetes and for weight management in people with overweight or obesity.[17–19] Semaglutide has also shown a reduction in risk of cardiovascular events in people with type 2 diabetes and elevated cardiovascular risk and a reduction in risk of clinically important kidney outcomes in people with type 2 diabetes and chronic kidney disease, both compared with placebo.[17,18,20] Effects of cagrilintide and semaglutide on blood glucose regulation might be partly mediated by their effects on bodyweight; however, there is evidence to suggest that the two molecules might also provide glycaemic benefits through distinct yet complementary mechanisms, with semaglutide increasing insulin secretion and cagrilintide potentially enhancing insulin sensitivity.[15,21,22] Furthermore, there is a need for additional therapies that can improve glycaemia while minimising hypoglycaemia and also provide cardiorenal benefits in individuals with type 2 diabetes treated with basal insulin, particularly given the high residual cardiorenal risk that persists in advanced type 2 diabetes despite intensive glycaemic management.[23] Both cagrilintide and semaglutide have shown weight-reducing effects with no additional risk of hypoglycaemia,[15] with semaglutide conferring additional cardiorenal benefits beyond weight reduction.[20,24] Therefore, the combination of cagrilintide and semaglutide could provide additional treatment benefits compared with those observed with the monocomponents alone. In REDEFINE 2, a phase 3, placebo-controlled clinical study, once per week cagrilintide 2·4 mg plus semaglutide 2·4 mg (hereafter cagrilintide–semaglutide [2·4 mg each]) was associated with clinically meaningful bodyweight reduction and statistically significant improvements in glycaemic control and other cardiometabolic risk factors in adults with type 2 diabetes and overweight or obesity.[25]

Cagrilintide–semaglutide is being assessed across the type 2 diabetes spectrum in the REIMAGINE clinical study programme.[26,27] The phase 3 study REIMAGINE 3 is the first to investigate an amylin receptor agonist and GLP-1 receptor agonist combination in people with type 2 diabetes inadequately controlled with basal insulin. We aimed to examine the effect of two doses of cagrilintide–semaglutide on change in glycated haemoglobin (HbA_{1c}) compared with matched placebo, as an add-on treatment in people with inadequately controlled type 2 diabetes receiving once per day basal insulin with or without metformin.

Methods

Study design

REIMAGINE 3 was a global, phase 3a, randomised, placebo-controlled, multicentre, parallel-group study that was double-blinded within dose levels and done at 46 centres (university hospitals, health-care centres, research centres, and other clinical trial sites) in the USA, China, Japan, Serbia, Slovakia, and South Africa. Complete lists of the REIMAGINE 3 investigators and study sites by country are provided in the appendix (pp 2–4).

See Online for appendix

The study was done in accordance with the principles of the Declaration of Helsinki and International Conference on Harmonisation Good Clinical Practice guidelines. The protocol (appendix p 64) was approved by an independent ethics committee or institutional review board at each study site (appendix pp 5–6). People with type 2 diabetes were not formally involved in the study design and conduct. The statistical analysis plan can be found in the appendix (p 64). This study was registered with ClinicalTrials.gov (NCT06323161) on March 21, 2024, and is complete.

Participants

Eligible participants were aged 18 years or above, had received a diagnosis of type 2 diabetes 180 days or more before screening, and were receiving a stable once per day dose of basal insulin (minimum of 0·25 U/kg per day or 20 U/day) alone or in combination with metformin (at effective or maximum tolerated dose) for 90 days before screening. Participants were required to have HbA_{1c} of 7·0–10·5% (53–91 mmol/mol) and BMI of 25 kg/m² or more at screening. Key exclusion criteria were renal impairment with an estimated glomerular filtration rate of less than 30 mL/min per 1·73 m², treatment with any medication for diabetes or obesity other than stated in the inclusion criteria within 90 days before screening, uncontrolled and potentially unstable diabetic retinopathy or maculopathy, known hypoglycaemia unawareness, and recurrent severe hypoglycaemic episodes within the past year. A full list of inclusion and exclusion criteria can be found in the appendix (pp 7–8). Sex at birth (male or female), race, and ethnicity were all self-reported to site staff, who recorded this information. All participants provided written informed consent before any study-related activities.

Randomisation and masking

Participants were centrally randomly assigned through the Randomisation and Trial Supply Management system or Interactive Web Response System. Sequence generation and treatment allocation were conducted by this same system, and treatment was dispensed by the investigator at study visits. After a screening period of up to 3 weeks, eligible participants were randomly assigned in a 2:2:1:1 ratio to receive once per week treatment with subcutaneous cagrilintide–semaglutide (2·4 mg each) or cagrilintide–semaglutide (1·0 mg each) or dose-matched placebo (2·4 mg plus 2·4 mg or 1·0 mg plus 1·0 mg) as an add-on therapy to once per day basal insulin, with or without metformin. We chose the 2·4 mg dose based on previously reported clinical data,[15] whereas the 1·0 mg dose was derived from modelling analyses (data not shown) and included to provide greater flexibility for treatment individualisation. Randomisation was stratified by HbA_{1c} of less than 8·5% at screening (yes or no) and country (Japan; yes or no). Participants, care providers, investigators, and outcome assessors were masked within each dose level to treatment allocation. Active treatments were visually identical to placebo, and participants were assigned unique identifiers for treatment dispensing to maintain masking of dispensing investigators to treatment allocation.

Procedures

Participants received treatment for 40 weeks followed by a 7-week follow-up period (appendix p 58). Treatment was initiated at 0·25 mg plus 0·25 mg once per week for 4 weeks, followed by stepwise dose escalation to 0·5 mg plus 0·5 mg, 1·0 mg plus 1·0 mg, 1·7 mg plus 1·7 mg, and 2·4 mg plus 2·4 mg every 4 weeks until the target dose was achieved. Participants assigned to the 1·0 mg each dose had an 8-week dose escalation followed by 32 weeks of maintenance treatment, and participants assigned to the 2·4 mg each dose had a 16-week dose escalation followed by 24 weeks of maintenance treatment. Protocol-specified strategies to mitigate and manage gastrointestinal adverse events during dose escalation and dose maintenance are summarised in the appendix (p 9). Dose reduction or delay of escalation was permitted if the current dose was associated with unacceptable adverse events or if a participant with a BMI in the lower normal range continued to lose weight and had an associated health concern, according to the investigator's discretion; dose reduction was required for participants reaching a BMI of less than 18·5 kg/m². Participants were allowed to continue in the study at the maximum tolerated dose and were recommended to attempt re-escalation to the target dose at least once. All treatments were self-administered into the thigh, abdomen, or upper arm, at any time of day irrespective of meals, with a pre-filled dual-chamber single-dose pen injector device. Rescue medicine could be offered as an add-on therapy to participants with persistent and unacceptable hyperglycaemia at the discretion of the investigator; locally approved glucose-lowering therapies could be prescribed in accordance with local guidelines and recommendations from the American Diabetes Association and European Association for the Study of Diabetes, except for GLP-1 receptor agonists, dipeptidyl peptidase-4 inhibitors, dual glucose-dependent insulinotropic polypeptide and GLP-1 receptor agonists, and amylin receptor agonists.

Participants continued to receive treatment with once per day basal insulin (insulin glargine [U-100, U-300, original branded, or any generic or biosimilar available], insulin detemir, insulin degludec, or neutral protamine Hagedorn insulin). Insulin titration was guided by fasting plasma glucose (FPG) concentrations, and symptoms of hypoglycaemia or hyperglycaemia. At randomisation, participants with a baseline HbA_{1c} of more than 8·0% (>64 mmol/mol) continued with their pre-study basal insulin dose and participants with a baseline HbA_{1c} of 8·0% or less (≤64 mmol/mol) had their pre-study basal insulin dose reduced by 20%. After randomisation, adjustment of basal insulin was reviewed on a weekly basis by the investigators, and titration was based on self-monitoring of blood glucose concentrations with a fasting glucose target of 80–130 mg/dL (4·4–7·2 mmol/L; appendix p 10). Participants on metformin were to maintain their pre-randomisation dose. Participants attended 12 on-site visits, scheduled to take place at screening, randomisation (week 0), weeks 4, 8, 12, 16, 20, 28, 36, and 40, and the end of study follow-up. Outcome measurements by week of study visits are shown in the appendix (p 11). Adverse events, including hypoglycaemic episodes, were assessed throughout the study. Further details on methods for patient-reported outcomes, including self-monitoring of blood glucose, are provided in the appendix (pp 12–13).

Outcomes

Overall, 29 prespecified efficacy outcomes were evaluated across two cagrilintide–semaglutide dose groups, comprising one primary endpoint, eight confirmatory secondary endpoints, and 20 supportive secondary endpoints. A full list of all prespecified primary and secondary objectives and endpoints is provided in the appendix (pp 14–15). The primary endpoint was mean change in HbA_{1c} (percentage points) from baseline to end of treatment at week 40. Confirmatory secondary endpoints were relative mean change in bodyweight (as a percentage) from baseline to week 40; a 10% or more bodyweight reduction at week 40; a 15% or more bodyweight reduction at week 40; HbA_{1c} less than 7·0% (<53 mmol/mol) at week 40; HbA_{1c} of 6·5% or less (≤48 mmol/mol) at week 40; mean changes from baseline to week 40 in FPG (mmol/L) and insulin dose (U); and discontinuation of insulin at week 40. We did post-hoc analyses by race for change in HbA_{1c} and relative change in bodyweight. We also did a preplanned exploratory analysis measuring absolute mean change in bodyweight (kg) from baseline to week 40 to complement the prespecified endpoint of relative change in bodyweight. Supportive safety endpoints were the numbers from baseline to the end of the study of treatment-emergent adverse events, clinically significant hypoglycaemic episodes (<3·0 mmol/L [54 mg/dL]), and severe hypoglycaemic episodes (hypoglycaemia associated with severe cognitive impairment requiring external assistance for recovery). An independent external event adjudication committee (appendix p 16) was established for ongoing masked adjudication of selected adverse events (malignant neoplasm, acute coronary syndrome, cerebrovascular events, heart failure, and coronary revascularisation procedure) and deaths.

Statistical analysis

A sample size of approximately 270 participants was considered sufficient to ensure 93% power to jointly substantiate the superiority hypotheses for both cagrilintide–semaglutide doses (2·4 mg each or 1·0 mg each) with respect to HbA_{1c} and bodyweight (98% for the 2·4 mg each dose). Assumptions made when determining power are listed in the appendix (p 17).

Efficacy endpoints were analysed in all participants randomly assigned to treatment (full analysis set). To address the primary and confirmatory secondary endpoints regarding change from baseline to week 40 in HbA_{1c} and change from baseline to week 40 in bodyweight, one-sided hypotheses were evaluated for the superiority of cagrilintide–semaglutide versus placebo at each dose

(HbA_{1c} ≥0 percentage points *vs* <0 percentage points; bodyweight ≥0% *vs* <0%). Data from the two dose-matched placebo groups were pooled and reported as one. Adjustment for multiplicity associated with the testing of the hypotheses was done through the weighted Bonferroni-based closed testing procedure.[28] The type I error was preserved in the strong sense at a nominal two-sided 5%. If the main hypotheses were substantiated, superiority testing for the remaining confirmatory secondary efficacy hypotheses was tested with a fixed sequence testing strategy (appendix pp 18–20). Overall, 16 hypotheses were formally tested within the multiplicity-controlled framework across both cagrilintide–semaglutide dose groups; discontinuation of insulin at week 40 was not tested as data were too sparse for analysis. Supportive secondary endpoints and additional analyses were not part of the confirmatory testing strategy.

Two observation periods were used for efficacy endpoints: the on-treatment without rescue period (from randomisation to 14 days after the last administration of trial study drug or initiation of rescue medication) and the in-study period (from randomisation to end of study). Two estimands (efficacy and treatment regimen; defined in the appendix p 21) were used to assess efficacy from different perspectives and account for intercurrent events and missing data differently.[29,30] For the efficacy estimand, the statistical analysis used a mixed model for repeated measurements that included treatment group, stratification, and region as factors and baseline HbA_{1c} as a continuous covariate, all nested within visit. For the treatment regimen estimand, the statistical analysis was a pattern mixture model with multiple imputation followed by an analysis of covariance model with treatment group, stratification, and region as factors and baseline HbA_{1c} as a continuous covariate. All reported results are for the efficacy estimand, unless stated otherwise. Further details on the analysis and imputation methods used to handle missing data, including statistical methods for exploratory and post-hoc analyses, are summarised in the appendix (pp 22–25). All results from statistical analyses were accompanied by two-sided p values and 95% CIs, with significance defined as $p<0·05$.

Safety endpoints were assessed in all participants exposed to at least one dose of the trial study drug (safety analysis set) and were analysed descriptively. For safety analyses, the in-study period was defined as for the efficacy analyses and was used to assess adverse event data with a long lag-time. The on-treatment period was defined as the time from randomisation to the date of last trial study drug administration plus 49 days or last study site contact, whichever came first, and was used to assess adverse event data with acute onset and hypoglycaemic episodes. Clinically significant hypoglycaemic episodes occurring while on treatment were confirmed by a blood glucose meter and were analysed based on the full analysis set.

No independent data monitoring committee was established for this study due to several aspects of the study design, including the short duration, the absence of a large or vulnerable population, and because the study was not designed as a long-term outcomes trial. Analyses were done with R (version 4.4.1).

Role of the funding source

The funder of the study was responsible for the study design, preparing the study protocol and statistical analysis plan, site monitoring, data collation, and performing the statistical analyses. The investigators were responsible for study-related medical decisions and data collection. All authors were responsible for data interpretation and contributed to the manuscript content. Medical writing and editorial support were paid for by the funder.

Results

Between March 26 and Nov 29, 2024, we screened 340 individuals for inclusion and 274 were randomly assigned to receive treatment with cagrilintide–semaglutide (2·4 mg each; n=90), cagrilintide–semaglutide (1·0 mg each; n=93), or matched placebo (n=91). All randomly assigned participants were exposed to treatment and included in the full analysis set and safety analysis set (figure 1). Most participants completed treatment (80 [89%] of 90 with cagrilintide–semaglutide [2·4 mg each]; 80 [86%] of 93 with cagrilintide–semaglutide [1·0 mg each]; and 87 [96%] of 91 with placebo) and completed the study (88 [98%] of 90 with cagrilintide–semaglutide [2·4 mg each], 87 [94%] of 93 with cagrilintide–semaglutide [1·0 mg each], and 90 [99%] of 91 with placebo). Overall, 72 (80%) of 90 participants in the cagrilintide–semaglutide (2·4 mg each) and 78 (84%) of 93 in the cagrilintide–semaglutide (1·0 mg each) dose groups reached the maximum dose during the trial (appendix p 26).

Baseline characteristics are presented in table 1 and in the appendix (pp 27–28). The mean age of the participants was 59 years (SD 10), 159 (58%) of 274 participants were male and 115 (42%) were female, 125 (46%) were Asian, and 129 (47%) were White. The mean HbA_{1c} was 8·8% (SD 1·0; 72·5 mmol/mol [SD 10·7]) and diabetes duration was 14·9 years (SD 7·6). The mean once per day basal insulin dose was 34 U (19) and 234 (85%) participants were receiving metformin. The mean bodyweight was 88·2 kg (SD 17·9) and BMI was 31·6 kg/m² (5·9).

Both doses of cagrilintide–semaglutide met the primary objective of superiority to placebo in mean change in HbA_{1c}. Using the efficacy estimand, the mean change in HbA_{1c} from baseline to week 40 was –2·33% (SE 0·08) with cagrilintide–semaglutide (2·4 mg each), –2·10% (SE 0·08) with cagrilintide–semaglutide (1·0 mg each), and –0·66% (SE 0·11) with placebo (figure 2; appendix pp 29–32). The estimated treatment difference

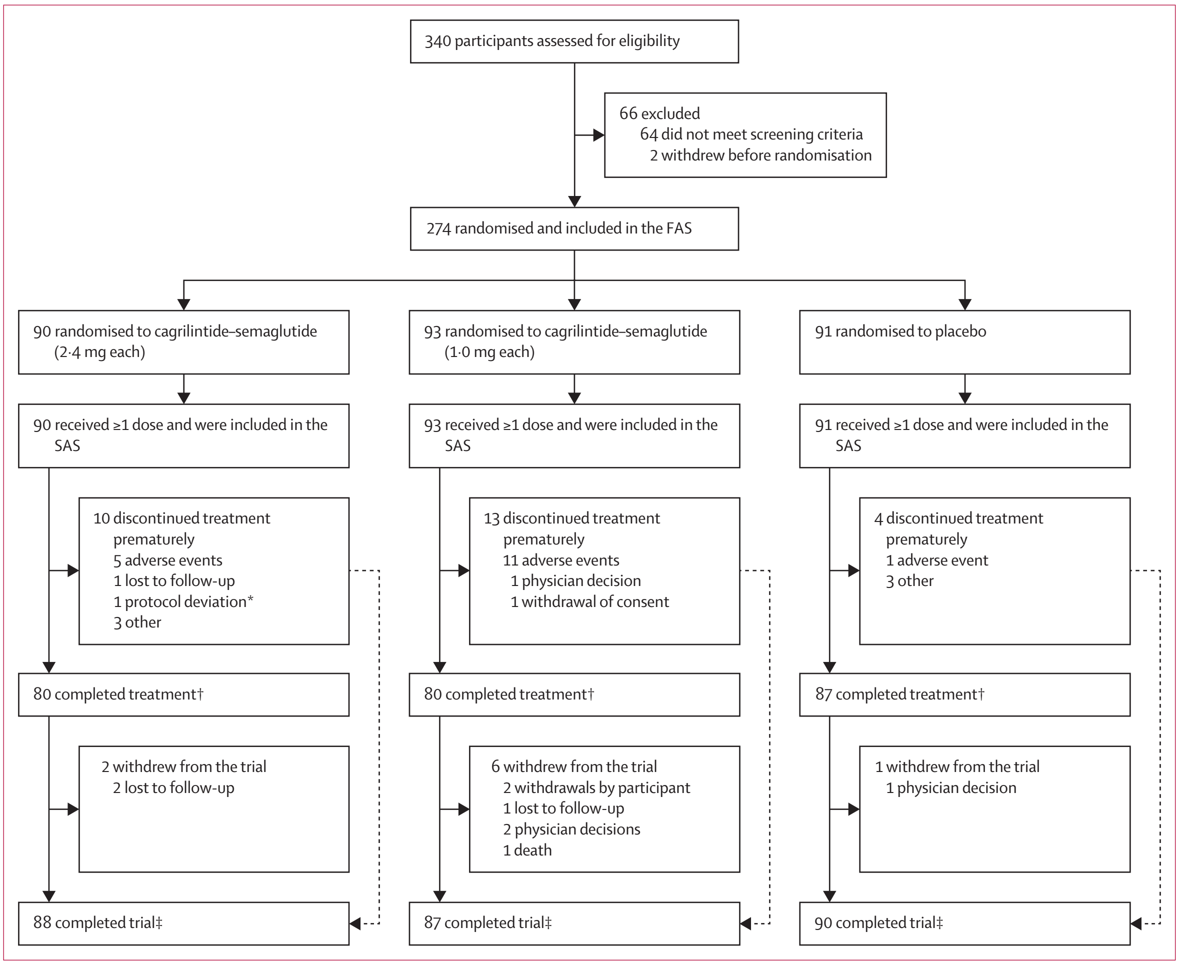

Figure 1: **Study profile**
SAS=safety analysis set. *Included in the study in violation of the inclusion and/or exclusion criteria. †Participants were considered to have completed treatment if they were on treatment at week 40. ‡Participants were considered to have completed the study if they attended the end-of-study visit at week 47, regardless of whether they completed treatment.

(ETD) was –1·68 percentage points (95% CI –1·95 to –1·41; p<0·0001) for cagrilintide–semaglutide (2·4 mg each) versus placebo and –1·44 percentage points (–1·71 to –1·17; p<0·0001) for cagrilintide–semaglutide (1·0 mg each) versus placebo. The observed mean HbA_{1c} at week 40 was less than 7·0% in both cagrilintide–semaglutide dose groups (figure 2A). Both cagrilintide–semaglutide doses were also superior to placebo for meeting HbA_{1c} targets, change in FPG, and change in once per day basal insulin dose. Significantly greater proportions of participants receiving cagrilintide–semaglutide than placebo met HbA_{1c} targets of less than 7·0% (ETD 59·2% [95% CI 51·8–66·6] for cagrilintide–semaglutide [2·4 mg each] and 53·7% [46·0–61·4] for cagrilintide–semaglutide [1·0 mg each]; both p<0·0001) and targets of 6·5% or less (54·2% [47·4–61·0] for cagrilintide–semaglutide [2·4 mg each] and 42·2% [35·1–49·3] for cagrilintide–semaglutide [1·0 mg each]; both p<0·0001; figure 2; appendix pp 29–32). At week 40, there were significant reductions in FPG versus placebo with cagrilintide–semaglutide (2·4 mg each; ETD –1·6 mmol/L [95% CI –2·2 to –1·0]; p<0·0001) and cagrilintide–semaglutide (1·0 mg each; –1·2 mmol/L [–1·9 to –0·5]; p=0·0013; figure 2; appendix pp 29–32). Both cagrilintide–semaglutide doses provided significantly greater reductions in mean once per day basal insulin dose versus placebo at week 40. The ETD versus placebo was –20 U (95% CI –24 to –16) with cagrilintide–semaglutide (2·4 mg each) and –20 U (95% CI –24 to –16) with cagrilintide–semaglutide (1·0 mg each; both p<0·0001; figure 2; appendix pp 29–32). The proportion of participants who were on-treatment without rescue medication at week 40 was greater with cagrilintide–semaglutide (2·4 mg each; 75 [83%] of 90) and cagrilintide–semaglutide (1·0 mg each; 78 [84%] of 93) compared with placebo (64 [70%] of 91). Rescue medications initiated during the study are summarised in the appendix (p 33). Results for all glycaemic primary and confirmatory secondary endpoints were similar with the treatment regimen estimand (appendix pp 34–37, 59–60).

Cagrilintide–semaglutide (2·4 mg each) and cagrilintide–semaglutide (1·0 mg each) were superior to placebo for change in bodyweight and meeting the 10% or more and 15% or more bodyweight reduction thresholds at week 40. Using the efficacy estimand, the relative mean change in bodyweight from baseline to week 40 was –12·0% (SE 0·7) with cagrilintide–semaglutide (2·4 mg each), –10·4% (SE 0·7) with cagrilintide–semaglutide (1·0 mg each), and 1·1% (SE 0·4) with placebo. The ETD versus placebo was –13·1% (95% CI –14·7 to –11·5; p<0·0001) for cagrilintide–semaglutide (2·4 mg each) and –11·6% (–13·2 to –9·9; p<0·0001) for cagrilintide–semaglutide (1·0 mg each; figure 3; appendix pp 29–32). Significantly greater proportions of participants receiving cagrilintide–semaglutide (2·4 mg each) had a bodyweight reduction of 10% or more (58% [50·7–64·5]; p<0·0001) and 15% or more (31% [25·3–37·1]; p<0·0001) compared with placebo (figure 3; appendix pp 29–32). Participants receiving cagrilintide–semaglutide (1·0 mg each) were also more likely than those in the placebo group to meet these targets. Results for all weight-related confirmatory secondary endpoints were similar with the treatment regimen estimand (appendix pp 34–37, 61–62).

Supportive secondary efficacy outcomes with the efficacy estimand are summarised in the appendix (pp 29–32). Both cagrilintide–semaglutide doses were associated with observed improvements in other glycaemic, weight-related, and cardiometabolic endpoints, including changes in mean 7-point self-monitoring of blood glucose, 7-point self-monitoring of blood glucose postprandial increment, and waist circumference (figure 3E); bodyweight reductions of 5% or more and 20% or more; and changes in systolic blood pressure, high-sensitivity C-reactive protein, and leptin. Cagrilintide–semaglutide (2·4 mg each) was also associated with improvements in lipid variables (total cholesterol, HDL cholesterol, non-HDL cholesterol, LDL

	Cagrilintide–semaglutide (2·4 mg each; n=90)	Cagrilintide–semaglutide (1·0 mg each; n=93)	Placebo (n=91)	Total (N=274)
Sex				
Female	38 (42%)	41 (44%)	36 (40%)	115 (42%)
Male	52 (58%)	52 (56%)	55 (60%)	159 (58%)
Age, years	59·2 (9·3)	58·8 (10·8)	59·0 (10·4)	59·0 (10·2)
Race				
Asian	43 (48%)	44 (47%)	38 (42%)	125 (46%)
Black or African American	3 (3%)	3 (3%)	8 (9%)	14 (5%)
White	41 (46%)	43 (46%)	45 (49%)	129 (47%)
Other*	3 (3%)	3 (3%)	0	6 (2%)
Hispanic or Latino ethnicity	11 (12%)	12 (13%)	16 (18%)	39 (14%)
Bodyweight, kg	88·9 (18·5)	86·7 (15·8)	88·9 (19·4)	88·2 (17·9)
BMI, kg/m²	32·0 (6·0)	31·4 (5·5)	31·5 (6·2)	31·6 (5·9)
Waist circumference, cm	106·8 (16·0)	104·9 (13·5)	104·9 (14·1)	105·5 (14·6)
HbA_{1c}, %	8·8 (0·9)	8·8 (1·0)	8·8 (1·0)	8·8 (1·0)
HbA_{1c}, mmol/mol	72·7 (10·1)	72·4 (10·5)	72·4 (11·4)	72·5 (10·7)
Duration of diabetes, years	15·1 (7·9)	14·6 (7·4)	14·9 (7·7)	14·9 (7·6)
FPG, mmol/L				
Mean (SD)	8·4 (2·3)†	8·2 (3·0)‡	8·8 (2·7)‡	8·5 (2·7)§
Median (IQR)	8·4 (6·8–9·7)†	7·7 (6·1–9·5)‡	8·1 (7·1–10·3)‡	8·1 (6·7–9·8)§
FPG, mg/dL				
Mean (SD)	151·9 (41·7)†	147·3 (53·3)‡	158·3 (48·0)‡	152·5 (48·0)§
Median (IQR)	151·2 (122·0–174·3)†	139·3 (109·0–170·6)‡	146·2 (127·2–185·1)‡	145·1 (120·0–176·8)§
Systolic blood pressure, mm Hg	128·1 (14·4)	131·6 (14·4)	133·4 (13·2)	131·1 (14·2)
Diastolic blood pressure, mm Hg	78·1 (9·6)	80·5 (9·7)	80·4 (8·0)	79·7 (9·2)
eGFR, mL/min per 1·73 m²	98·6 (16·3)	92·7 (20·8)	97·4 (16·9)	96·2 (18·2)
Once per day basal insulin dose at baseline, U				
Mean (SD)	31·6 (14·9)	35·8 (22·8)	34·7 (17·6)	34·0 (18·8)
Median (IQR)	26 (20–36)	28 (22–40)	28 (22–42)	28 (22–40)
Metformin at screening	81 (90%)	76 (82%)	77 (85%)	234 (85%)

Data are mean (SD) or n (%), unless otherwise specified. Percentages might not sum to 100 as a result of rounding. eGFR=estimated glomerular filtration rate. FPG=fasting plasma glucose. HbA_{1c}=glycated haemoglobin. *American Indian, Alaska Native, or not reported. †n=89. ‡n=90. §n=269.

Table 1: **Baseline demographics and clinical characteristics**

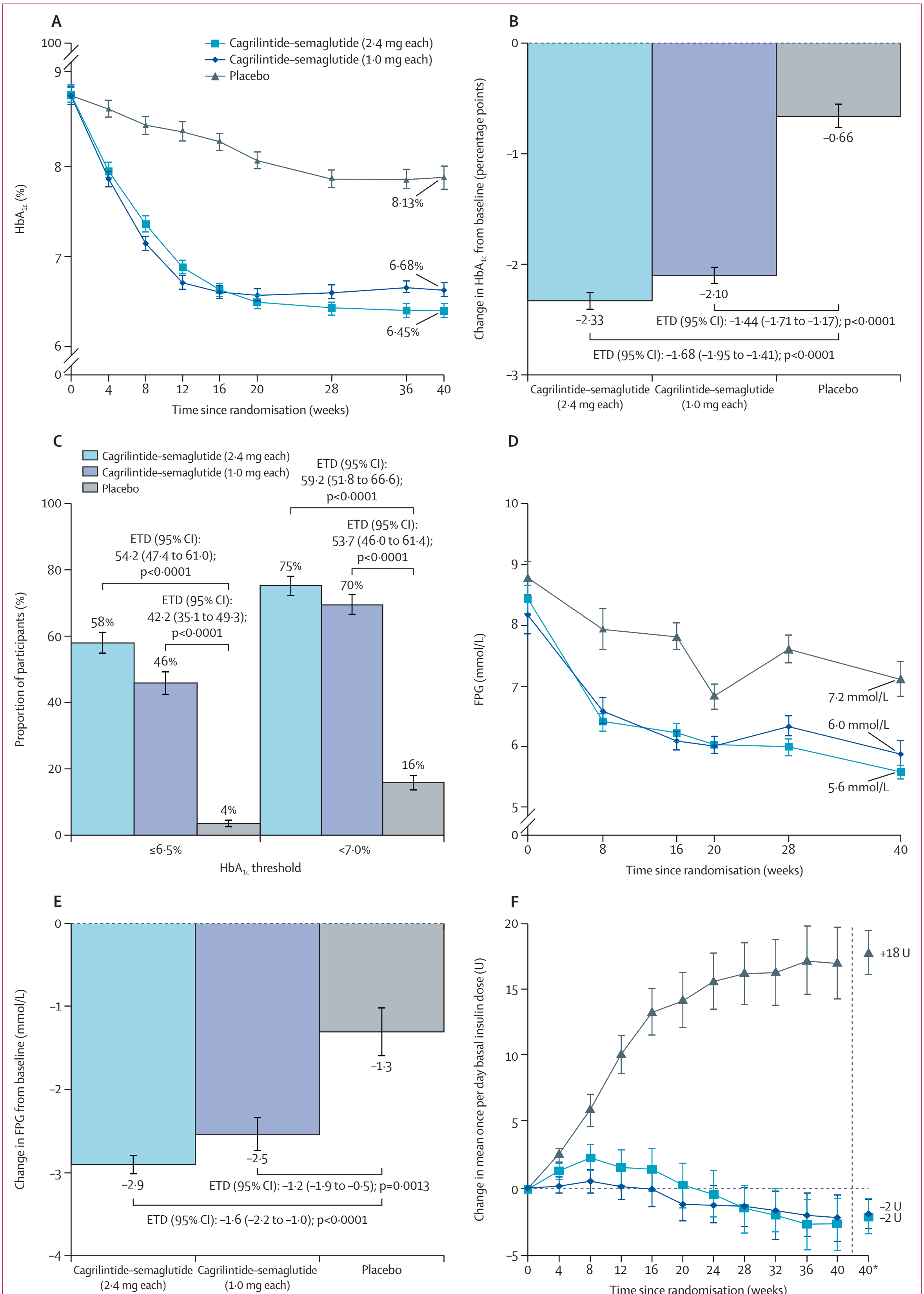

***Figure 2:* Change from baseline in glycaemic endpoints (efficacy estimand)**
(A) Mean HbA_{1c} (%) over time (overall mean HbA_{1c} at baseline was 8·8% [SD 1·0] or 72·5 mmol/mol [SD 10·7]). (B) Mean change in HbA_{1c} (percentage points) from baseline to week 40. (C) Proportions of participants meeting HbA_{1c} targets of 6·5% or less and less than 7·0% at week 40. (D) Mean FPG (mmol/L) over time (overall mean FPG at baseline was 8·5 mmol/L [SD 2·7]; median 8·1 mmol/L [IQR 6·7–9·8]). (E) Mean change in FPG (mmol/L) from baseline to week 40. (F) Mean change in once per day basal insulin dose (U) over time (overall mean once per day basal insulin dose at baseline was 34 U [SD 19]; median 28 U [IQR 22–40]). Data are for the full analysis set and efficacy estimand. Error bars in panels A, D, and F indicate SE of the mean. Error bars in panels B, C, and E indicate SE. ETD=estimated treatment difference. FPG=fasting plasma glucose. HbA_{1c}=glycated haemoglobin. *Estimated mean at week 40.

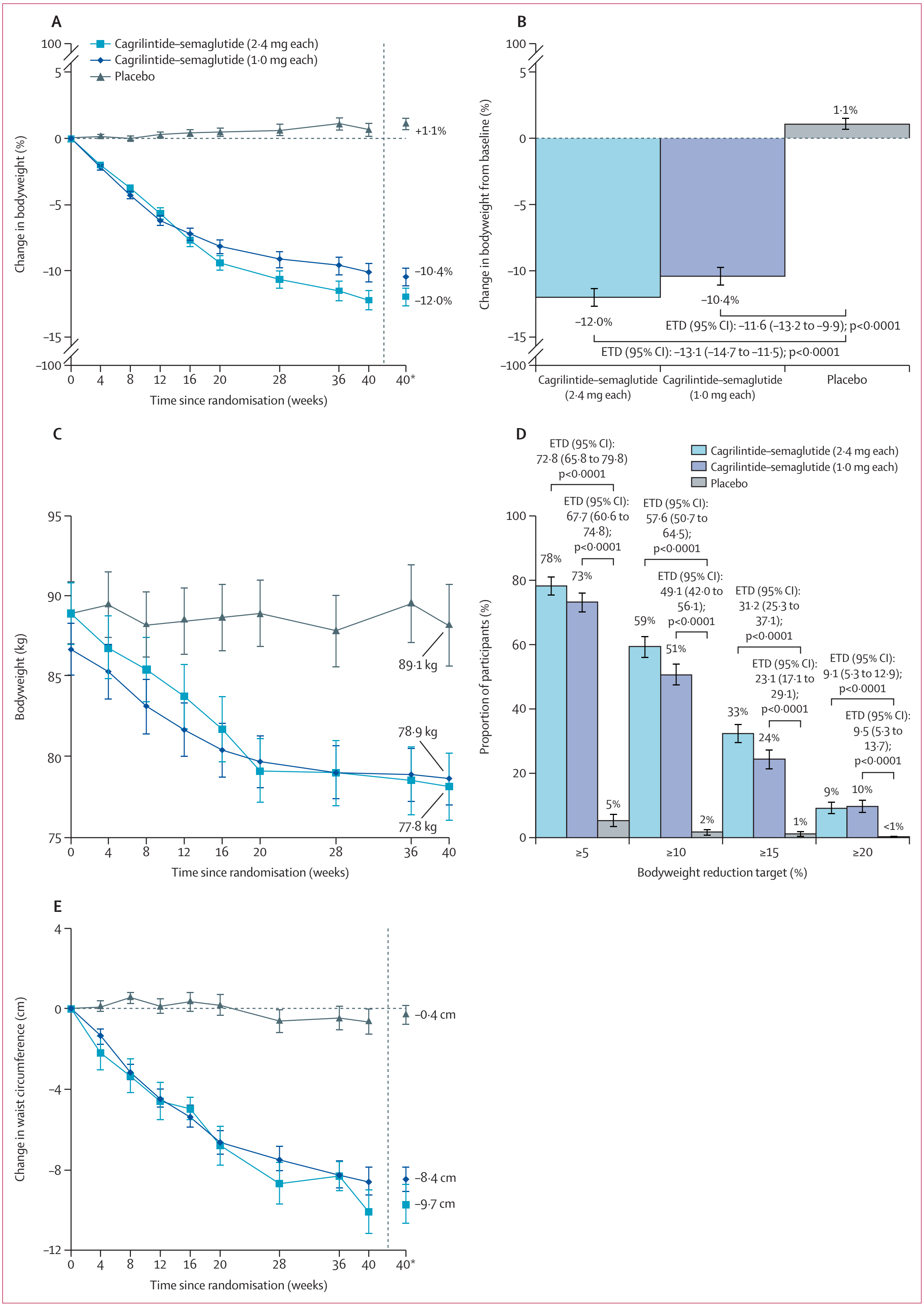

***Figure 3:* Change from baseline in bodyweight-related endpoints (efficacy estimand)**
(A) Mean relative change in bodyweight (%) over time (overall mean bodyweight at baseline was 88·2 kg [SD 17·9]). (B) Mean relative change in bodyweight (%) from baseline to week 40. (C) Mean absolute bodyweight (kg) over time. (D) Proportions of participants who had bodyweight reductions of 5% or more, 10% or more, 15% or more, and 20% or more at week 40; proportions of participants with 5% or more and 20% or more reduction were not adjusted for multiplicity. (E) Mean change in waist circumference (cm) over time; this endpoint was not adjusted for multiplicity (overall mean waist circumference at baseline was 105·5 cm [SD 14·6]). Data are for the full analysis set and efficacy estimand. Error bars in panels A, C, and E indicate SE of the mean. Error bars in panels B and D indicate SE. ETD=estimated treatment difference. *Estimated mean at week 40.

	Cagrilintide–semaglutide (2·4 mg each; n=90)			Cagrilintide–semaglutide (1·0 mg each; n=93)			Placebo (n=91)		
	n (%)	Number of events	Event rate per 100 years	n (%)	Number of events	Event rate per 100 years	n (%)	Number of events	Event rate per 100 years
Adverse events and serious adverse events									
Any adverse event*	72 (80%)	396	492·7	66 (71%)	280	340·2	65 (71%)	224	273·9
Serious adverse event*	5 (6%)	10	12·4	7 (8%)	7	8·5	8 (9%)	9	11·0
Adverse event leading to trial product discontinuation*	6 (7%)	8	10·0	11 (12%)	13	15·8	1 (1%)	1	1·2
Gastrointestinal-related adverse events leading to trial product discontinuation	2 (2%)	3	4·0	8 (9%)	9	12·0	0	0	0
Fatal events*	0	0	0	1 (1%)†	1	1·2	0	0	0
Hypoglycaemic episodes‡									
Level 1 (alert value)	40 (44%)	163	216·2	35 (38%)	137	183·0	41 (45%)	205	260·8
Level 2 (clinically significant)	13 (14%)	23	30·5	8 (9%)	10	13·4	13 (14%)	32	40·7
Level 3 (severe)	0	0	0	0	0	0	0	0	0
Level 2 or 3	13 (14%)	23	30·5	8 (9%)	10	13·4	13 (14%)	32	40·7
Nocturnal level 2 or 3	5 (6%)	8	10·6	1 (1%)	1	1·3	3 (3%)	5	6·4
Selected safety focus areas									
Gastrointestinal disorders§	51 (57%)	148	196·3	42 (45%)	117	156·3	21 (23%)	32	40·7
Retinal disorders*§	13 (14%)	16	19·9	5 (5%)	6	7·3	10 (11%)	12	14·7
Neoplasms*§	7 (8%)	8	10·0	6 (6%)	10	12·1	2 (2%)	3	3·7
Hypersensitivity reactions§	4 (4%)	4	5·3	3 (3%)	3	4·0	3 (3%)	3	3·8
Gallbladder-related disorders§	4 (4%)	6	8·0	0	0	0	0	0	0
Injection site reactions§	1 (1%)	1	1·3	0	0	0	0	0	0
Malignant neoplasms*¶	1 (1%)	1	1·2	2 (2%)	2	2·4	1 (1%)	1	1·2
Most frequent adverse events by system order class and preferred term‖									
Gastrointestinal disorders									
Nausea	28 (31%)	42	55·7	25 (27%)	39	52·1	3 (3%)	3	3·8
Diarrhoea	14 (16%)	22	29·2	10 (11%)	16	21·4	3 (3%)	3	3·8
Vomiting	13 (14%)	31	41·1	9 (10%)	16	21·4	1 (1%)	1	1·3
Constipation	10 (11%)	11	14·6	11 (12%)	13	17·4	5 (5%)	5	6·4
Abdominal discomfort	5 (6%)	5	6·6	2 (2%)	2	2·7	3 (3%)	3	3·8
Metabolism and nutrition disorders									
Decreased appetite	18 (20%)	23	30·5	14 (15%)	17	22·7	1 (1%)	2	2·5
Hypoglycaemia	6 (7%)	6	8·0	2 (2%)	2	2·7	3 (3%)	3	3·8
Eye disorders									
Diabetic retinopathy	8 (9%)	8	10·6	4 (4%)	4	5·3	8 (9%)	9	11·5
General disorders and administration site conditions									
Fatigue	7 (8%)	9	11·9	3 (3%)	5	6·7	3 (3%)	3	3·8
Infections and infestations									
Nasopharyngitis	7 (8%)	8	10·6	10 (11%)	14	18·7	12 (13%)	16	20·4
Nervous system disorders									
Headache	6 (7%)	10	13·3	5 (5%)	6	8·0	3 (3%)	4	5·1
Investigations									
Blood glucose increased	0	0	0	0	0	0	5 (5%)	8	10·2

Data are from the safety analysis set from the on-treatment period, unless otherwise stated. *In-study period. †One death was reported in the cagrilintide–semaglutide (1·0 mg each) group; the event adjudication committee established the cause of death as malignancy (exocrine and endocrine pancreatic cancer). ‡Episodes with onset date within the on-treatment period; hypoglycaemic episodes were defined according to the American Diabetes Association 2018 classification: level 1 (blood glucose <3·9 mmol/L and ≥3·0 mmol/L); level 2 (blood glucose <3·0 mmol/L); level 3 (requiring assistance from another person for recovery). §Based on a pre-defined Medical Dictionary for Regulatory Activities (version 28.1) search. ¶The malignant neoplasms recorded here are those that were confirmed by the event adjudication committee. ‖Adverse events occurring in 5% or more of participants in any treatment group.

Table 2: **Selected adverse events in the safety analysis set**

cholesterol, very LDL cholesterol, and triglycerides). Consistent findings were found with the treatment regimen estimand (appendix pp 34–37). In post-hoc analyses, change in HbA_{1c} and relative change in bodyweight did not differ across racial groups (appendix pp 38–39).

Adverse events collected during the in-study period were reported by 72 (80%) of 90 participants receiving cagrilintide–semaglutide (2·4 mg each), 66 (71%) of 93 receiving cagrilintide–semaglutide (1·0 mg each), and 65 (71%) of 91 receiving placebo (table 2). A full list of adverse events with onset during the on-treatment period is included in the appendix (pp 40–54). Gastrointestinal disorders were the most common category of adverse events, occurring in 51 (57%) of 90 participants receiving cagrilintide–semaglutide (2·4 mg each), 42 (45%) of 93 receiving cagrilintide–semaglutide (1·0 mg each), and 21 (23%) of 91 receiving placebo. Most gastrointestinal adverse events were mild or moderate in severity with onset during the dose escalation period. Other selected safety focus areas are presented in table 2.

Serious adverse events collected during the in-study period were reported by five (6%) of 90 participants in the cagrilintide–semaglutide (2·4 mg each) group, seven (8%) of 93 in the cagrilintide–semaglutide (1·0 mg each) group, and eight (9%) of 91 in the placebo group (table 2). Adverse events leading to treatment discontinuation were reported by six (7%) of 90 participants in the cagrilintide–semaglutide (2·4 mg each) group, 11 (12%) of 93 in the cagrilintide–semaglutide (1·0 mg each) group, and one (1%) of 91 in the placebo group. Gastrointestinal disorders were the cause of permanent discontinuation for two (2%) of 90 participants in the cagrilintide–semaglutide (2·4 mg each) group and eight (9%) of 93 in the cagrilintide–semaglutide (1·0 mg each) group. Serious adverse events and adverse events leading to permanent treatment discontinuation are reported by system order class and preferred term in the appendix (pp 55–57). One death was reported in the cagrilintide–semaglutide (1·0 mg each) group; the external and independent event adjudication committee determined the cause of death as malignancy. There were no deaths in the cagrilintide–semaglutide (2·4 mg each) and placebo groups.

No severe hypoglycaemic episodes (level 3) were reported. Level 2 hypoglycaemic episodes occurred in similar proportions of participants across treatment groups, affecting 13 (14%) of 90 participants in the cagrilintide–semaglutide (2·4 mg each) group, eight (9%) of 93 in the cagrilintide–semaglutide (1·0 mg each) group, and 13 (14%) of 91 in the placebo group. The numbers of participants who had level 1 hypoglycaemic episodes were also similar across treatment groups (table 2).

Discussion

In REIMAGINE 3, cagrilintide–semaglutide (2·4 mg each) and cagrilintide–semaglutide (1·0 mg each) provided superior reductions in HbA_{1c} compared with placebo, with both cagrilintide–semaglutide groups having an observed mean HbA_{1c} of less than 7·0% at week 40 from mean baseline HbA_{1c} of 8·8%. Improvements in glycaemic control were accompanied by robust bodyweight reductions and improvements in other cardiometabolic variables, without any additional risk of hypoglycaemia and no severe hypoglycaemic episodes in this study. In contrast, participants in the placebo group had a modest reduction in HbA_{1c} by up titrating their once per day basal insulin dose and an increase in bodyweight. REIMAGINE 3 is the first study to report findings from the combination of a long-acting amylin receptor agonist and a GLP-1 receptor agonist in a population with type 2 diabetes inadequately controlled with basal insulin with or without metformin. The results of this study complement previous studies and contribute additional evidence for cagrilintide–semaglutide in type 2 diabetes.[15,26,27]

In terms of glycaemic control, cagrilintide–semaglutide (2·4 mg each) and cagrilintide–semaglutide (1·0 mg each) were associated with significant reductions in HbA_{1c}. In addition, significantly higher proportions of participants treated with cagrilintide–semaglutide met HbA_{1c} targets of less than 7·0% and 6·5% or less compared with placebo, despite a mean baseline HbA_{1c} of 8·8%. These findings are particularly meaningful in a population with long-standing advanced type 2 diabetes inadequately controlled on basal insulin, in whom conventional treatment intensification is often limited not only by increased risks of hypoglycaemia, but also by increasing therapeutic complexity.[4,31]

The improvements in glycaemic control with cagrilintide–semaglutide were accompanied by robust reductions in bodyweight of 10–12% versus an increase in bodyweight of 1·1% with placebo. Significant proportions of participants in both cagrilintide–semaglutide groups also met bodyweight reduction targets of 10% or more and 15% or more. The magnitude of bodyweight reduction seen with cagrilintide–semaglutide over the 40-week study period is notable in a population with basal insulin-treated type 2 diabetes that is predisposed to weight gain,[32] especially given the relatively low mean baseline bodyweight of 88·2 kg. Furthermore, steady-state weight had not been achieved by week 40, suggesting the potential for further bodyweight reduction over longer time periods.

In both cagrilintide–semaglutide groups, there was an estimated reduction of 20 U in once per day basal insulin requirements from baseline to week 40 compared with placebo. The absolute change from baseline was modest (–2 U) in the cagrilintide–semaglutide (2·4 mg each) group, which might reflect protocol-related constraints on insulin management, including the absence of proactive dose reduction upon achievement of glycaemic targets and the high upper fasting glucose target threshold of 130 mg/dL for dose adjustment. Use of HbA_{1c} targets to guide down titration might have enabled greater insulin dose reductions. Notably, participants in the placebo group increased their once per day basal insulin dose by 18 U over the course of the study with the same protocol-guided titration algorithm.

Semaglutide has been shown to reduce cardiovascular risk and risk of clinically important kidney outcomes in people with type 2 diabetes.[20,24] Treatment with cagrilintide–semaglutide (2·4 mg each) in the present study was associated with beneficial effects in other cardiometabolic risk factors, including systolic blood pressure, high-sensitivity C-reactive protein, lipids (total cholesterol, HDL cholesterol, non-HDL cholesterol, LDL cholesterol, very LDL cholesterol, and triglycerides), and leptin. These findings are consistent with the effects of cagrilintide–semaglutide seen in previous studies in people with type 2 diabetes.[15,25] The cardiovascular safety of cagrilintide–semaglutide (2·4 mg each) and its efficacy on cardiovascular outcomes in people living with established cardiovascular disease and overweight or obesity, with or without type 2 diabetes, are currently under investigation in the phase 3 study REDEFINE 3 (NCT05669755).[33]

The safety profile of cagrilintide–semaglutide was consistent with that observed in previous studies.[15,25–27] The most common adverse events were gastrointestinal symptoms that were mainly mild to moderate in severity; this is typical of GLP-1-based therapies, with these adverse events usually occurring during dose escalation and reducing over time.[34] Overall, there were no unexpected findings observed for the prespecified safety focus areas and no unexpected findings from additional safety areas and laboratory assessments. There were no severe hypoglycaemic events, and the proportion of participants experiencing non-severe clinically significant hypoglycaemia events was similar across treatment groups.

The consistent tolerability profile observed across both doses supports a scalable treatment approach, whereby the 1·0 mg each dose might be a clinically appropriate dose for some individuals, or could serve as an intermediary step before escalation to the full therapeutic dose of 2·4 mg each when additional glycaemic efficacy or weight reduction is required or desirable. Indeed, greater reductions in HbA_{1c} were seen in the higher cagrilintide–semaglutide dose group, whereas reductions in basal insulin requirement and the proportions of participants not requiring rescue medication were similar across dose groups, suggesting that some treatment effects might plateau at lower doses. Interpretation of these findings should consider baseline imbalances between cagrilintide–semaglutide dose groups, including modest differences in bodyweight and once per day basal insulin dose, which could have contributed to variability in glycaemic outcomes. Collectively, these observations support a role for individualised dosing when balancing expected efficacy against the additional titration steps to reach higher doses.

This study has several strengths. The study had high rates of treatment completion and study completion. It also had a greater representation of east Asian participants than similar studies, allowing for generalisability across the represented populations. Inclusion of east Asian participants is relevant given the characteristic contribution of β-cell dysfunction early in the type 2 diabetes disease trajectory in this population.[35] However, since most Asian participants were recruited at sites in China and Japan, the applicability of the study's findings in the context of broader Asian populations remains uncertain. Although most participants were treated with insulin glargine at baseline, the use of multiple basal insulin formulations reflects contemporary clinical practice and the international scope of the trial. Limitations include the small sample size and relatively short 40-week study duration, which did not permit assessment of long-term efficacy or safety. In addition, the study did not include an active comparator to advance basal insulin therapy with an alternative GLP-1-based treatment; however, this reflects the regulatory objective of the study to compare cagrilintide–semaglutide with conventional optimisation of basal insulin through further titration. Another potential limitation of this study is that fasting glucose titration targets were based on American Diabetes Association guidance; the use of lower fasting targets, as observed in other basal insulin treat-to-target studies,[36] might have resulted in greater reductions in HbA_{1c}. In this blinded, placebo-controlled trial, basal insulin adjustment was protocol-directed and primarily reactive to observed hypoglycaemia, which might not fully reflect the more individualised insulin down titration that could occur in clinical practice when initiating cagrilintide–semaglutide; this context should be considered when interpreting the hypoglycaemia findings from this study. Post-hoc subgroup analyses by race were done for change in HbA_{1c} and relative change in bodyweight; however, other endpoints were not assessed by race. Accordingly, the generalisability of findings across other endpoints should be interpreted with caution. Finally, the current trial was not designed to assess the weight-independent effects of cagrilintide–semaglutide; this could be explored in future longer-term studies in this population.

In conclusion, both cagrilintide–semaglutide doses (2·4 mg each and 1·0 mg each) had statistically superior reductions in HbA_{1c} versus placebo in people with type 2 diabetes who had inadequate glycaemic control on once per day basal insulin with or without metformin. These HbA_{1c} reductions were accompanied by robust reductions in bodyweight and improvements in other cardiometabolic variables. The safety and tolerability profile of cagrilintide–semaglutide was consistent with that of the GLP-1 receptor agonist drug class and previous safety data for cagrilintide, and no increase in the risk of hypoglycaemia was observed despite concomitant insulin use. These findings are of particular clinical relevance in a population for whom further treatment intensification beyond basal insulin is frequently limited by weight gain, hypoglycaemia risk, and increasing therapeutic complexity—barriers that remain largely unaddressed by conventional basal insulin escalation strategies. Cagrilintide–semaglutide therefore represents a promising therapeutic option for individuals with type 2 diabetes who have inadequate glycaemic control on basal insulin therapy, integrating the established benefits of semaglutide with the complementary effects of cagrilintide to address glycaemic control and bodyweight in this challenging-to-treat population.

Contributors

JR, NBJ, KRK, UKT, JBB, and ABJ contributed to the design, protocol, and/or conduct of the study. UKT was responsible for the statistical analyses. JR and UKT directly accessed and verified the underlying data reported in the manuscript. All authors participated in the interpretation of the data and critical review of the manuscript, approved the final and previous versions of the manuscript, had full access to all the data in the study, and had final responsibility for the decision to submit for publication.

Declaration of interests

JR reports clinical research grants from Amgen, Applied Therapeutics, AstraZeneca, Biomea Fusion, Boehringer Ingelheim, Corcept Therapeutics, Corxel, Eli Lilly, Hanmi, Merck, Novo Nordisk, Pfizer, Regeneron, Regor, Roche, Sanofi, Structure Therapeutics, and Terns; serves or has served on scientific advisory boards and received honorarium or consulting fees from Amgen, Applied Therapeutics, Arrowhead, Biomea Fusion, Boehringer Ingelheim, Corcept Therapeutics, Eccogene, Eli Lilly, Hanmi, i2O Therapeutics, Novo Nordisk, Regeneron, Roche, Sanofi, Structure Therapeutics, and Terns; and has received honoraria for lectures from Eli Lilly, Novo Nordisk, and Sanofi. LKB has received consulting honoraria from Amgen, Bayer, Dexcom, Eli Lilly, Endogenex, Novo Nordisk, Pfizer, Sanofi, and Xeris. RG is an employee and shareholder at Novo Nordisk. FG reports research grants from Eli Lilly and Roche Diabetes Care; consulting fees from Eli Lilly and Novo Nordisk; support for attending meetings or travel from AstraZeneca, Boehringer Ingelheim, Eli Lilly, LifeScan, Medtronic, Merck Sharp & Dohme, Novo Nordisk, Roche Diabetes Care, and Sanofi Aventis; participation on advisory boards for AstraZeneca, Boehringer Ingelheim, Eli Lilly, LifeScan, MedImmune, Medtronic, Merck Sharp & Dohme, Novo Nordisk, Roche Diabetes Care, and Sanofi Aventis; unpaid leadership for the European Foundation for the Study of Diabetes and European Association for the Study of Diabetes, Società Italiana di Endocrinologia, and Fo.Ri.SIE; and support for medical writing and statistical analysis from AstraZeneca, Eli Lilly, Novo Nordisk, and Sanofi Aventis. NBJ is an employee and shareholder at Novo Nordisk. KRK has received advisory or consulting fees support from Antag Therapeutics, Metsera, Novo Nordisk, Roche Pharmaceuticals, and vTv Therapeutics; research-related contracts (paid to the institution) from Bayer, Boehringer Ingelheim, Carmot, Diasome, Eli Lilly, GentiBio, Novo Nordisk, Rhythm Pharmaceuticals, Roche Pharmaceuticals, and vTv Therapeutics; and salary support through the National Institutes of Health (K12TR004416, U54DK118612, and R21AA031892). UKT is an employee and shareholder at Novo Nordisk. JBB has received salary support from clinical trial contracts to his employer by Corcept Therapeutics, Dexcom, GentiBio, Novo Nordisk, and Sparrow Pharmaceuticals; has received salary support from consulting contracts to his employer by Novo Nordisk; is a consultant with personal compensation from Aardvark Therapeutics, Altimmune, Alveus Therapeutics, Amgen, Antag, Aqua Medical, AstraZeneca, Boehringer Ingelheim, Corcept Therapeutics, Dexcom, Eli Lilly, embecta, General Medicines Inc, GentiBio, Kayothera, Metsera, MiniMed, Pfizer, Recordati, Roche, Sparrow Pharmaceuticals, Vertex, vTv Therapeutics, and Zealand; has stock or stock options in Aardvark Therapeutics, Glyscend, Mellitus Health, Metsera, Pendulum Therapeutics, Praetego, and Stability Health; and has received salary support from grants from the National Institutes of Health, Breakthrough T1D, and PCORI. ABJ has received clinical trials funding from Abbott, Amgen, Argenx, AstraZeneca, Eli Lilly, Novo Nordisk, Roche, and Sanofi Aventis; has received speaking honorarium or advisory compensation from Abbott, AbbVie, Amgen, Antibody, AstraZeneca, Bausch Healthcare, Bayer, Boehringer Ingelheim, Celltrion, Dexcom, Diabetes Canada, Eli Lilly, embecta, GSK, HLS Therapeutics, Insulet, Janssen, Kailera, Medtronic, Merck, Moderna, Novartis, Novo Nordisk, Pendopharm, Pfizer, Roche, Sandoz, Takeda, and Ypsomed.

Data sharing

An authorised researcher can request access to clinical study data by submitting a research proposal for review and approval by Novo Nordisk and an internal independent review panel. An authorised researcher is an external researcher who can show the legitimacy and scientific merit of their request. Requests are usually considered after the research is finished and the main results have been published. If the research supports a regulatory application, requests will be considered after the product and its intended use are approved in both the EU and the USA. Participants' clinical data will be anonymised, following an approved internal process, before data are shared to external third parties. For details on how to request access to clinical data, visit https://www.novonordisk-trials.com.

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